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Self-winding hose reels for pressure washers

Abstract

An assembly for high-pressure cleaning and a method for manufacturing an assembly for high-pressure cleaning are provided. The assembly includes a self-winding hose reel, a pressure washer, and a fastening mechanism. The pressure washer includes a pump and a frame. The frame includes a first side pipe and a second side pipe parallel to the first side pipe. The fastening mechanism is designed to secure the self-winding hose reel to the frame between the first side pipe and the second side pipe. The fastening mechanism allows rotation of the self-winding hose reel with respect to a plane formed by the first side pipe and the second side pipe.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4135669	12/1978	Bridges	239/391	B05B 9/007
4238074	12/1979	Coons	N/A	N/A
4700422	12/1986	Russell	134/168C	B08B 9/045
4700737	12/1986	Nelson	N/A	N/A
4773113	12/1987	Russell	15/4	E03F 9/005
5419497	12/1994	Warrington	239/722	A62C 27/00
5653584	12/1996	Mazzucato	417/364	F04B 53/00
5980220	12/1998	Mazzucato	417/364	F04B 1/148
6761135	12/2003	Becktold	N/A	N/A
6935642	12/2004	Craig	280/47.24	F04B 17/06
8398097	12/2012	Junk	417/364	B08B 3/026
8915231	12/2013	Raasch	123/363	F02D 31/00
10888887	12/2020	Masters	N/A	N/A
D925845	12/2020	Alexander	D32/15	N/A
D932717	12/2020	Xu	D32/15	N/A
2006/0076779	12/2005	Craig	290/1A	F02B 63/04
2009/0272412	12/2008	Steinhauser et al.	N/A	N/A
2011/0253229	12/2010	Mattson	137/355.16	B05B 7/0093
2012/0018534	12/2011	Gilpatrick	239/310	B05B 7/0876
2012/0241537	12/2011	Schwei	239/147	B05B 9/007
2014/0119949	12/2013	Wischstadt	417/34	F04B 17/05
2015/0102121	12/2014	Dey	239/69	B05B 9/01
2016/0167869	12/2015	Gilpatrick	222/251	F04B 53/18
2017/0304873	12/2016	Luby et al.	N/A	N/A
2018/0021800	12/2017	Janik	239/146	B05B 15/62
2018/0133729	12/2017	Masters	N/A	A47L 11/4091
2019/0299236	12/2018	Rancourt	N/A	F04B 17/05
2020/0001313	12/2019	Bindl	N/A	B05B 12/006
2020/0023399	12/2019	Gordon	N/A	B05B 9/007
2020/0171522	12/2019	Sugiura	N/A	N/A
2021/0078049	12/2020	Nushart	N/A	E01H 1/101

2022/0339655	12/2021	Wuerfl	N/A	B05B 9/0861
2024/0240442	12/2023	Sphar	N/A	E03C 1/302

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
MU90014812	12/2011	BR	N/A
2019206395	12/2018	WO	N/A

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Background/Summary

TECHNICAL FIELD

(1) This disclosure relates to pressure washers. More specifically, this disclosure relates to self-winding hose reels for pressure washers.

BACKGROUND

(2) Pressure washer systems operate by propelling pressurized water and are widely used for cleaning various surfaces to effectively remove dirt, grime, and contaminants from the surfaces. Pressure washers usually have an engine acting as a power source for driving the pressure washer, a pump for pressurizing the water, a frame for providing structural support and protection for components of the pressure washer, a spray nozzle for controlling the shape and intensity of a water stream, and a high-pressure water hose that connects a pump outlet to the spray nozzle. Pressure washers may have other components depending on the specific model and type of a pressure washer.

(3) The high-pressure water hose serves as a conduit through which pressurized water is delivered from the pump to the spray nozzle. Conventional pressure washers, for example, models widely sold in the North American market, have loose high-pressure water hoses that hang on a side or in the middle of handles of the pressure washers. Users have to wrap and unwrap the high-pressure water hoses by hand. Since the high-pressure water hoses are very stiff, the users may experience difficulties when handling the high-pressure water hoses.

(4) Some conventional pressure washers, for example, models provided on the European market, have a manual hose reel on them. The manual hose reel may be connected to a frame of the pressure washer and used for convenient storage and management of the high-pressure water hoses when not in use. Operating the manual hose reel requires physical effort to manually crank a handle to wind or unwind the high-pressure water hose. This can become tiresome for users, especially if the high-pressure water hose is long and heavy.

SUMMARY

(5) This summary is provided to introduce a selection of concepts in a simplified form that are further described in the Detailed Description below. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

(6) According to an example embodiment of the present disclosure, an assembly for high-pressure cleaning is provided. The assembly may include a self-winding hose reel, a pressure washer, and a fastening mechanism. The pressure washer may include a pump and a frame. The frame may include a first side pipe and a second side pipe parallel to the first side pipe. The fastening mechanism may be designed to secure the self-winding hose reel to the frame between the first side pipe and the second side pipe.

(7) According to another embodiment of the present disclosure, a method for manufacturing an assembly for high-pressure cleaning is provided. The method may commence with providing a self-winding hose reel. The method may proceed with providing a pressure washer. The pressure washer may include a pump and a frame. The frame may include a first side pipe and a second side pipe parallel to the first side pipe. The method may proceed with providing a fastening mechanism designed to secure the self-winding hose reel to the frame between the first side pipe and the second side pipe.

(8) Other example embodiments of the disclosure and aspects will become apparent from the following description taken in conjunction with the following drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) Exemplary embodiments are illustrated by way of example and not limitation in the figures of the accompanying drawings, in which like references indicate similar elements.

(2) FIG. 1 is a front perspective view of an assembly for high-pressure cleaning with a vertical shaft engine powered pressure washer and a self-winding hose reel, according to an example embodiment.

(3) FIG. 2 is a front perspective view of an assembly for high-pressure cleaning with a self-winding hose reel in a horizontal position, according to an example embodiment.

(4) FIG. 3A is a rear perspective view of an assembly for high-pressure cleaning with a self-winding hose reel in a horizontal position, according to an example embodiment.

(5) FIG. 3B is a detailed view of a fastening mechanism, according to an example embodiment.

(6) FIG. 4 is a general view of a self-winding hose reel connected to a fastening mechanism, according to an example embodiment.

(7) FIG. 5 is a side cross-sectional view of a self-winding hose reel, according to an example embodiment.

(8) FIG. 6 is an expanded view of a self-winding hose reel, according to an example embodiment.

(9) FIG. 7A is a front view of a self-winding hose reel, according to an example embodiment.

(10) FIG. 7B is a detailed view of a pawl of a self-winding hose reel, according to an example embodiment.

(11) FIG. 8A illustrates rotation of a drum to pull a water hose out of a self-winding hose reel, according to an example embodiment.

(12) FIG. 8B illustrates rotation of a drum to retract a water hose into a self-winding hose reel, according to an example embodiment.

(13) FIG. 9 is a front perspective view of an assembly for high-pressure cleaning with a horizontal shaft engine powered pressure washer, a self-winding hose reel, and a vertical type handle, according to an example embodiment.

(14) FIG. 10 is a rear view of an assembly for high-pressure cleaning with a horizontal shaft engine powered pressure washer, a self-winding hose reel, and a vertical type handle, according to an example embodiment.

(15) FIG. 11 is a front perspective view of an assembly for high-pressure cleaning with a horizontal shaft engine powered pressure washer, a self-winding hose reel, and a horizontal type handle, according to an example embodiment.

(16) FIG. 12 is a rear perspective view of an assembly for high-pressure cleaning with a horizontal shaft engine powered pressure washer, a self-winding hose reel, and a horizontal type handle, according to an example embodiment.

(17) FIG. 13 is a left side view of an assembly for high-pressure cleaning with a horizontal shaft engine powered pressure washer, a self-winding hose reel, and a horizontal type handle, according

to an example embodiment.

(18) FIG. 14 is a top view of an assembly for high-pressure cleaning with a horizontal shaft engine powered pressure washer, a self-winding hose reel, and a horizontal type handle, according to an example embodiment.

(19) FIG. 15 is a flow chart of a method for manufacturing an assembly for high-pressure cleaning, according to an example embodiment.

DETAILED DESCRIPTION

(20) The following detailed description of embodiments includes references to the accompanying drawings, which form a part of the detailed description. Approaches described in this section are not prior art to the claims and are not admitted to be prior art by inclusion in this section. The drawings show illustrations in accordance with example embodiments. These example embodiments, which are also referred to herein as “examples,” are described in enough detail to enable those skilled in the art to practice the present subject matter. The embodiments can be combined, other embodiments can be utilized, or structural, logical, and operational changes can be made without departing from the scope of what is claimed. The following detailed description is, therefore, not to be taken in a limiting sense, and the scope is defined by the appended claims and their equivalents.

(21) Generally, the embodiments of this disclosure relate to assemblies for high-pressure cleaning. An assembly for high-pressure cleaning may include a pressure washer, a self-winding hose reel, and a fastening mechanism for the self-winding hose reel. The pressure washer may include a pump and a frame. The frame may have a first side pipe and a second side pipe parallel to the first side pipe. The fastening mechanism may be designed to secure the self-winding hose reel to the frame between the first side pipe and the second side pipe.

(22) The self-winding hose reel may include a water hose, a drum onto which the water hose is wound, a spring disposed in the drum, and a housing for accommodating the components of the self-winding hose reel. The spring actuating the rotation of the drum allows the water hose to be automatically wound up inside the housing of the self-winding hose reel after use.

(23) The pressure washer may include any type of pressure washer available on the market. In example embodiments, the pressure washer may include a pressure washer powered by a vertical shaft engine and a pressure washer powered by a horizontal shaft engine. The present disclosure provides several example embodiments for attaching the self-winding hose reel to these types of pressure washers.

(24) In the vertical shaft engine powered pressure washer, a pump is installed under a vertical shaft engine, i.e., on the bottom of the pressure washer. For this type of pressure washers, the self-winding hose reel may be arranged over the vertical shaft engine. The self-winding hose reel may be attached by the fastening mechanism to a handle of the pressure washer. The handle may be a part of a frame of the pressure washer. The fastening mechanism may be configured to enable the self-winding hose reel to rotate between a horizontal position and a vertical position. Specifically, the self-winding hose reel may fold down into the horizontal position during the operation of the pressure washer and may fold up into the vertical position when access to a fuel tank or other elements of the pressure washer is needed.

(25) In the horizontal shaft engine powered pressure washer, the pump is installed beside a horizontal shaft engine. There are two structures of the pressure washers powered by horizontal shaft engines: a vertical style pressure washer and a horizontal style pressure washer. The vertical style pressure washer powered by a horizontal shaft engine has bigger wheels on the rear side of the pressure washer and a vertical type handle. For this type of pressure washer, the self-winding hose reel is arranged under the horizontal shaft engine, i.e., on the bottom of the pressure washer, and is attached by the fastening mechanism to a frame of the pressure washer.

(26) The horizontal style pressure washer powered by a horizontal shaft engine has smaller wheels on the front side of the pressure washer and a horizontal type handle. For this type of pressure

washer, the self-winding hose reel is arranged on the horizontal type handle.

(27) Referring now to the drawings, various embodiments are described in which like reference numerals represent like parts and assemblies throughout the several views. It should be noted that the reference to various embodiments does not limit the scope of the claims attached hereto. Additionally, any examples outlined in this specification are not intended to be limiting and merely set forth some of the many possible embodiments for the appended claims.

(28) FIG. 1 is a front perspective view of an assembly for high-pressure cleaning shown as an assembly **100**, according to an example embodiment. The assembly **100** may include a self-winding hose reel **102**, a pressure washer **104**, and a fastening mechanism **106** designed to secure the self-winding hose reel **102** to the pressure washer **104**. The pressure washer **104** may include one of the following: an electrically powered high-pressure washer, a gasoline engine powered high-pressure washer, and any other type of pressure washer.

(29) The pressure washer **104** may include a frame **110**. The frame **110** may provide structural support and protection for the components of the pressure washer **104**. The frame **110** may include a first side pipe **112** and a second side pipe **114** parallel to the first side pipe **112**. The fastening mechanism **106** may be configured to secure the self-winding hose reel **102** to the frame **110** between the first side pipe **112** and the second side pipe **114**. The frame **110** may further include a handle **116** connecting the first side pipe **112** and the second side pipe **114**. A user may use the handle **116** to navigate the assembly **100**.

(30) The assembly **100** shown in FIG. 1 includes the pressure washer **104** powered by a vertical shaft engine **118**. Therefore, the assembly **100** is a vertical shaft engine powered pressure washer with a self-winding hose reel. The pressure washer **104** may further include a pump **108**. The pump **108** may be located under the vertical shaft engine **118**. For example, the pump **108** may be arranged on an axis of a shaft of the vertical shaft engine **118**. In the example embodiment shown in FIG. 1 and FIG. 2, the self-winding hose reel **102** may be secured above the vertical shaft engine **118**.

(31) The fastening mechanism **106** may allow rotation of the self-winding hose reel **102** with respect to a plane **120** formed by the first side pipe **112** and the second side pipe **114**. Specifically, the fastening mechanism **106** may be configured to enable the self-winding hose reel **102** to rotate between a vertical position and a horizontal position. FIG. 1 shows the assembly **100** with the self-winding hose reel **102** in a vertical position, in which the self-winding hose reel **102** is secured in the plane **120** formed by the first side pipe **112** and the second side pipe **114**. The self-winding hose reel **102** may be folded up into the vertical position when access to components (e.g., a fuel tank) of the pressure washer located below the self-winding hose reel **102** is needed.

(32) The assembly **100** may have other components required for the operation of the assembly **100**, such as a spray nozzle **122** configured to control the shape and intensity of the water stream, a trigger gun **124** for activating the flow of pressurized water from the pump **108** to the spray nozzle **122**, wheels **126** for easy maneuverability and transport, and so forth.

(33) The fastening mechanism **106** may further include a first pin hole (shown as a first pin hole **306** in FIG. 3B), a second pin hole **128**, a third pin hole **130**, and a fourth pin hole **132**. The pin holes are described in more detail below with reference to FIG. 3A and FIG. 3B.

(34) FIG. 2 is a front perspective view of the assembly **100** with the self-winding hose reel **102** in a horizontal position, according to an example embodiment. The self-winding hose reel **102** may be folded down from the vertical position shown in FIG. 1 into the horizontal position shown in FIG. 2 using the fastening mechanism **106**. The self-winding hose reel **102** may be in the horizontal position during the operation of the pressure washer **104**. In the example embodiment, the self-winding hose reel **102** may be arranged on the frame **110** to extend a water hose located in the self-winding hose reel **102** in the direction outwards from a front side of the assembly **100**, as shown by an arrow **202** in FIG. 2.

(35) The fastening mechanism **106** may be configured to secure the self-winding hose reel **102** with

respect to the plane **120** formed by the first side pipe **112** and the second side pipe **114** at a pre-determined non-zero angle **204**. In an example embodiment, the pre-determined non-zero angle **204** at which the self-winding hose reel **102** is secured with respect to the plane **120** is about 90 degrees. When the self-winding hose reel **102** is secured at the pre-determined non-zero angle **204**, the self-winding hose reel **102** may be arranged over the vertical shaft engine **118**.

(36) In an example embodiment, when the self-winding hose reel **102** is secured at the pre-determined non-zero angle **204**, the self-winding hose reel **102** may be arranged parallel to the pump **108**.

(37) FIG. **3A** is a rear perspective view of the assembly **100** with the self-winding hose reel **102** in a horizontal position, according to an example embodiment. FIG. **3B** is a detailed view A of the fastening mechanism **106**, according to an example embodiment.

(38) As shown in FIG. **3A** and FIG. **3B**, the fastening mechanism **106** may include a first plate **302** mounted to the first side pipe **112** and a second plate **304** mounted to the second side pipe **114**. The first plate **302** may include a first pin hole **306** and a second pin hole **128** (shown in FIG. **1**). The second plate **304** may include a third pin hole **130** (shown in FIG. **1**) and a fourth pin hole **132** (shown in FIG. **1**).

(39) Two position limiters may be located on each plate to ensure the right vertical position and the right horizontal position of the self-winding hose reel **102**. Specifically, as shown in FIG. **3B**, the first plate **302** may include a first limiter **310** and a second limiter **312**. Similarly, as shown in FIG. **3A**, the second plate **304** may include a third limiter **314** and a fourth limiter **316**. The first limiter **310** and the third limiter **314** may be designed to ensure a first position of the self-winding hose reel **102** in the plane **120** formed by the first side pipe **112** and the second side pipe **114**.

Specifically, upon rotating the self-winding hose reel **102** by the fastening mechanism **106** from a horizontal position into a vertical position in which the self-winding hose reel **102** is located in the plane **120**, the first limiter **310** and the third limiter **314** may restrict further movement of the fastening mechanism **106**. Therefore, the first limiter **310** and the third limiter **314** may prevent the self-winding hose reel **102** from further rotation and keep the self-winding hose reel **102** within the plane **120**.

(40) The second limiter **312** and the fourth limiter **316** may be designed to ensure a second position of the self-winding hose reel **102** at the pre-determined non-zero angle **204** with respect to the plane **120** formed by the first side pipe **112** and the second side pipe **114**. Specifically, upon rotating the self-winding hose reel **102** by the fastening mechanism **106** from a vertical position to a horizontal position, in which the self-winding hose reel **102** is located at the pre-determined non-zero angle **204** with respect to the plane **120**, the second limiter **312** and the fourth limiter **316** may restrict further movement of the fastening mechanism **106**. Therefore, the second limiter **312** and the fourth limiter **316** may prevent the self-winding hose reel **102** from further rotation and keep the self-winding hose reel **102** at the pre-determined non-zero angle **204** with respect to the plane **120**.

(41) FIG. **4** is a general view of the self-winding hose reel **102** connected to the fastening mechanism **106**, according to an example embodiment. The self-winding hose reel **102** may have a housing **406** enclosing the components of the self-winding hose reel **102**. The fastening mechanism **106** may further include two spring loaded latch pins located on both sides of the housing **406** of the self-winding hose reel **102**. Specifically, the fastening mechanism **106** may include a first spring loaded latch pin **402** attached to one side of the housing **406** of the self-winding hose reel **102** and a second spring loaded latch pin **404** attached to the other side of the housing **406** of the self-winding hose reel **102**. When the self-winding hose reel **102** is secured in the plane **120** formed by the first side pipe **112** and the second side pipe **114** (i.e., the self-winding hose reel **102** is in a vertical position as shown in FIG. **1**), the first spring loaded latch pin **402** may be engaged with the first pin hole **306** and the second spring loaded latch pin **404** may be engaged with the third pin hole **130**. When the self-winding hose reel **102** is secured at the pre-determined non-zero angle **204**

(i.e., the self-winding hose reel **102** is in a horizontal position as shown in FIG. 2), the first spring loaded latch pin **402** may be engaged with the second pin hole **128** (shown in FIG. 1) and the second spring loaded latch pin **404** may be engaged with the fourth pin hole **132** (shown in FIG. 1). (42) The fastening mechanism **106** may be configured to rotate the self-winding hose reel **102** around an axis **412** from a vertical position used for storage of the assembly **100** by about 90 degrees to a horizontal position used for operation of the assembly **100**. To actuate rotation of the self-winding hose reel **102**, the fastening mechanism **106** may include a first button **408** and a second button **410**. The first button **408** may be designed to release the first spring loaded latch pin **402** from either the first pin hole **306** or the second pin hole **128**. The second button **410** may be designed to release the second spring loaded latch pin **404** from either the third pin hole **130** or the fourth pin hole **132**.

(43) The first button **408** and the second button **410** may be clicked or pinched to retract the first spring loaded latch pin **402** out of the first pin hole **306** and/or the second pin hole **128** and retract the second spring loaded latch pin **404** out of the third pin hole **130** and/or the fourth pin hole **132**. In an example embodiment, the self-winding hose reel **102** may be in a vertical position with the first spring loaded latch pin **402** engaged with the first pin hole **306** and the second spring loaded latch pin **404** engaged with the third pin hole **130**. Upon clicking the first button **408** and the second button **410**, the first spring loaded latch pin **402** may be retracted out of the first pin hole **306** and the second spring loaded latch pin **404** may be retracted out of the third pin hole **130**. Then, the housing **406** of the self-winding hose reel **102** may be rotated clockwise by about 90 degrees. Upon rotation of the housing **406**, the first button **408** and the second button **410** may be released, thereby causing the first spring loaded latch pin **402** to engage with the second pin hole **128** and causing the second spring loaded latch pin **404** to engage with the fourth pin hole **132** to secure the self-winding hose reel **102** in a horizontal position for operation of the self-winding hose reel **102**.

(44) FIG. 5 is a side cross-sectional view of the self-winding hose reel **102**, according to an example embodiment. Components of the self-winding hose reel **102** are further shown in detail in FIG. 6.

(45) FIG. 6 is an expanded view of the self-winding hose reel **102**, according to an example embodiment. As shown in FIGS. 4, 5, 6, the housing **406** of the self-winding hose reel **102** (shown in FIG. 4) may include a front housing part **416** and a rear housing part **418**. The self-winding hose reel **102** may further include a drum **506** disposed inside the front housing part **416** and a sleeve shaft **508** fixed in the front housing part **416**. The self-winding hose reel **102** may further include a water intake **420** and water outlet **414** with a stop block **422** arranged on the housing **406**. The self-winding hose reel **102** may further include a water hose **512** wrapped on the drum **506**. The stop block **422** may be arranged around the water hose **512** to prevent the water hose **512** from being retracted completely into the housing **406**. The self-winding hose reel **102** may further include a first spring **516** placed around the sleeve shaft **508** inside the drum **506**. In an example embodiment, the first spring **516** may include a coil spring, a volute spring, and any other applicable type of the spring. When pulling out the water hose **512**, the drum **506** rotates and compresses the first spring **516** around the sleeve shaft **508**, so that the compressed first spring **516** can rotate the drum **506** to automatically retract the water hose **512**.

(46) The self-winding hose reel **102** may further include a first side disc **518** placed onto the sleeve shaft **508** on one side of the drum **506** and a second side disc **520** placed onto the sleeve shaft **508** on the other side of the drum **506**. The first side disc **518** and the second side disc **520** may enclose the space in which the water hose **512** can be wound on the drum **506**.

(47) FIG. 7A is a front view of the self-winding hose reel **102**, according to an example embodiment. The self-winding hose reel **102** may further have a pawl **702**.

(48) FIG. 7B is a detailed view of the pawl **702** of the self-winding hose reel **102**, according to an example embodiment. The pawl **702** may be mounted on a bracket **708** with a pin **704**. The bracket

708 may be welded on the sleeve shaft **508**. The pawl **702** may be pre-loaded by a second spring **706** at the end, so that the spring-loaded pawl **702** can swivel in a predetermined range. A first teathed edge **710** and a second teathed edge **712** (shown in FIG. 7A) may be arranged on the drum **506** for engagement with the pawl **702**.

(49) FIG. 8A illustrates rotation of the drum **506** to pull the water hose **512** out of the self-winding hose reel **102**, according to an example embodiment. FIG. 8B illustrates rotation of the drum **506** to retract the water hose **512** into the self-winding hose reel **102**, according to an example embodiment.

(50) As shown in FIG. 8A, when pulling out the water hose **512**, the drum **506** rotates clockwise (as shown by a first arrow **802**) and the first spring **516** is compressed. The pawl **702** may include a pawl face portion **810** and a pawl back portion **812**. The pawl **702** skids over the first teathed edge **710** and the second teathed edge **712** of the drum **506** smoothly. When stopping pulling out the water hose **512**, the drum **506** rotates counterclockwise slightly by the force of the compressed first spring **516** until the pawl **702** engages, by the pawl face portion **810**, one tooth on the first teathed edge **710** or the second teathed edge **712** to block the rotation.

(51) To activate the automatic retraction of the water hose **512**, the water hose **512** may be further pulled out for a predetermined distance until the pawl **702** reaches either a first teeth-free area **806** or a second teeth-free area **808**. The first teeth-free area **806** may be separated from the first teathed edge **710** by a first edge **814** and separated from the second teathed edge **712** by a second edge **816**. The second teeth-free area **808** may be separated from the first teathed edge **710** by a third edge **818** and separated from the second teathed edge **712** by a fourth edge **820**.

(52) As shown in FIG. 8B, during the counterclockwise rotation of the drum **506** (as shown by a second arrow **804**), the pawl back portion **812** skids over the first teathed edge **710** and the second teathed edge **712**, which does not block the rotation of the drum **506**, and the compressed first spring **516** retracts the water hose **512** automatically.

(53) FIG. 9 and FIG. 10 illustrate an assembly **900** for high-pressure cleaning, according to an example embodiment. FIG. 9 is a front perspective view of the assembly **900** for high-pressure cleaning. FIG. 10 is a rear view of the assembly **900** for high-pressure cleaning.

(54) The assembly **900** may include a self-winding hose reel **102**, a pressure washer **104**, and a fastening mechanism **106** designed to secure the self-winding hose reel **102** to the pressure washer **104**.

(55) The pressure washer **104** may include a frame **110**. The frame **110** may provide structural support and protection for the components of the pressure washer **104**. The frame **110** may include a first side pipe **112** and a second side pipe **114** parallel to the first side pipe **112**. The fastening mechanism **106** may be configured to secure the self-winding hose reel **102** to the frame **110** between the first side pipe **112** and the second side pipe **114**. The frame **110** may further include a handle **116** connecting the first side pipe **112** and the second side pipe **114**. A user may use the handle **116** to navigate the assembly **900**.

(56) The assembly **900** is a vertical style pressure washer, in which a handle **116** of the frame **110** is a vertical type handle, i.e., extends vertically. The pressure washer **104** may be powered by a horizontal shaft engine **902**. Therefore, the assembly **900** is a horizontal shaft engine powered pressure washer with a self-winding hose reel and a vertical type handle.

(57) The pressure washer **104** may further include a pump **108**. The pump **108** may be located beside the horizontal shaft engine **902**. For example, the pump **108** may be located on an axis of a shaft of the horizontal shaft engine **902**.

(58) In the example embodiment shown in FIG. 9 and FIG. 10, the self-winding hose reel **102** may be secured under the horizontal shaft engine **902**, i.e., on the bottom of the pressure washer **104**. The self-winding hose reel **102** may be attached by the fastening mechanism **106** to the frame **110** of the pressure washer **104**.

(59) In the example embodiment, the frame **110** may include a platform **904** arranged between the

first side pipe **112** and the second side pipe **114**. The horizontal shaft engine **902** may be arranged on a first surface **906** of the platform **904** and the self-winding hose reel **102** may be arranged on a second surface **908** of the platform **904** and below the horizontal shaft engine **902**. The first surface **906** may be opposite to the second surface **908** of the platform **904**. The pump **108** may be mounted on the shaft of the horizontal shaft engine **902**. The pump **108** may not touch the platform **904**.

(60) The self-winding hose reel **102** may be arranged on the frame **110** to extend a water hose located in the self-winding hose reel **102** in the direction outwards from a front side of the assembly **900**, as shown by an arrow **910** in FIG. **9**.

(61) FIGS. **11-14** illustrate an assembly **1100** for high-pressure cleaning, according to an example embodiment. FIG. **11** is a front perspective view of the assembly **1100** for high-pressure cleaning. FIG. **12** is a rear perspective view of the assembly **1100** for high-pressure cleaning. FIG. **13** is a left side view of the assembly **1100** for high-pressure cleaning. FIG. **14** is a top view of the assembly **1100** for high-pressure cleaning.

(62) The assembly **1100** may include a self-winding hose reel **102**, a pressure washer **104**, and a fastening mechanism **106** designed to secure the self-winding hose reel **102** to the pressure washer **104**.

(63) The pressure washer **104** may include a frame **110**. The frame **110** may provide structural support and protection for the components of the pressure washer **104**. The frame **110** may include a first side pipe **112** and a second side pipe **114** parallel to the first side pipe **112**. The fastening mechanism **106** may be configured to secure the self-winding hose reel **102** to the frame **110** between the first side pipe **112** and the second side pipe **114**. The frame **110** may further include a handle **116** connecting the first side pipe **112** and the second side pipe **114**. A user may use the handle **116** to navigate the assembly **1100**.

(64) The assembly **1100** is a horizontal style pressure washer, in which the handle **116** of the frame **110** is a horizontal type handle, i.e., extends horizontally. The pressure washer **104** may be powered by a horizontal shaft engine **902**. Therefore, the assembly **1100** is a horizontal shaft engine powered pressure washer with a self-winding hose reel and the horizontal type handle.

(65) The pressure washer **104** may further include a pump **108**. The pump **108** may be located beside the horizontal shaft engine **902**. For example, the pump **108** may be located on an axis of a shaft of the horizontal shaft engine **902**.

(66) The first side pipe **112** of the frame **110** may include a horizontal first side pipe part **1102**, a vertical first side pipe part **1104**, and an inclined first side pipe part **1106**. The second side pipe **114** of the frame **110** may include a horizontal second side pipe part **1202**, a vertical second side pipe part **1204**, and an inclined second side pipe part **1206**. The inclined first side pipe part **1106** and the inclined second side pipe part **1206** of the frame **110** may be connected by a frame connecting part **1108**. The inclined first side pipe part **1106**, the inclined second side pipe part **1206**, and the frame connecting part **1108** may constitute the handle **116** of the assembly **1100**.

(67) In an example embodiment, the fastening mechanism **106** may be connected to the handle **116**. For example, one side of the fastening mechanism **106** may be connected to a point of connection of the vertical first side pipe part **1104** and the inclined first side pipe part **1106**, and the other side of the fastening mechanism **106** may be connected to the point of connection of the vertical second side pipe part **1204** and the inclined second side pipe part **1206**. In an example embodiment, the self-winding hose reel **102** may be disposed on the handle **116** parallel to the inclined first side pipe part **1106** and the inclined second side pipe part **1206**.

(68) In an example embodiment, the self-winding hose reel **102** may be disposed on the handle **116** parallel to the inclined first side pipe part **1106** and the inclined second side pipe part **1206**. The assembly **1100** may further have a first bracket **1112** for connecting the self-winding hose reel **102** to the inclined first side pipe part **1106** of the frame **110**. The assembly **1100** may further have a second bracket **1114** for connecting the self-winding hose reel **102** to the inclined second side pipe

part **1206** of the frame **110**.

(69) The self-winding hose reel **102** is arranged on the horizontal type handle **116** to extend a water hose located in the self-winding hose reel **102** in the direction outwards from a rear side of the assembly **1100**, as shown by an arrow **1110** in FIG. **11**. In the assembly **1100** shown in FIG. **11**, the water hose is extended in a direction opposite to the direction of extension of water hoses in the assembly **100** shown in FIG. **2** and the assembly **900** shown in FIG. **9**.

(70) FIG. **15** is a flow chart of a method **1500** for manufacturing an assembly for high-pressure cleaning, according to an example embodiment. In some embodiments, the operations may be combined, performed in parallel, or performed in a different order. The method **1500** may also include additional or fewer operations than those illustrated.

(71) In block **1502**, the method **1500** may commence with providing a self-winding hose reel. In block **1504**, the method **1500** may proceed with providing a pressure washer. In an example embodiment, the pressure washer may include an electrically powered high-pressure washer or a gasoline engine powered high-pressure washer. The pressure washer may include a pump and a frame. The frame may include a first side pipe and a second side pipe parallel to the first side pipe. In an example embodiment, assembly may further include an engine. The self-winding hose reel may be secured above the engine.

(72) In block **1506**, the method **1500** may proceed with providing a fastening mechanism designed to secure the self-winding hose reel to the frame between the first side pipe and the second side pipe. The fastening mechanism may enable rotation of the self-winding hose reel with respect to a plane formed by the first side pipe and the second side pipe. The fastening mechanism may be configured to secure the self-winding hose reel with respect to the plane formed by the first side pipe and the second side pipe at a pre-determined non-zero angle. In an example embodiment, when the self-winding hose reel is secured at the pre-determined non-zero angle, the self-winding hose reel may be arranged parallel to the pump.

(73) In an example embodiment, the fastening mechanism may include a first plate mounted to the first side pipe and a second plate mounted to the second side pipe. The first plate may include a first pin hole and a second pin hole. The second plate may include a third pin hole and a fourth pin hole. The fastening mechanism may further include a first spring loaded latch pin and a second spring loaded latch pin. The first spring loaded latch pin may be attached to a housing of the self-winding hose reel. The second spring loaded latch pin may be attached to the housing of the self-winding hose reel. When the self-winding hose reel is secured in the plane formed by the first side pipe and the second side pipe, the first spring loaded latch pin may be engaged with the first pin hole and the second spring loaded latch pin may be engaged with the third pin hole. When the self-winding hose reel is secured at the pre-determined non-zero angle, the first spring loaded latch pin may be engaged with the second pin hole and the second spring loaded latch pin may be engaged with the fourth pin hole.

(74) The first plate may include a first limiter and a second limiter. The second plate may include a third limiter and a fourth limiter. The first limiter and the third limiter may be configured to ensure a first position of the self-winding hose reel in the plane formed by the first side pipe and the second side pipe. The second limiter and the fourth limiter may be configured to ensure a second position of the self-winding hose reel at the pre-determined non-zero angle with respect to the plane formed by the first side pipe and the second side pipe.

(75) The fastening mechanism may further include a first button and a second button. The first button may be configured to release the first spring loaded latch pin from either the first pin hole or the second pin hole. The second button may be configured to release the second spring loaded latch pin from either the third pin hole or the fourth pin hole.

(76) In an example embodiment, the frame may further include a platform arranged between the first side pipe and the second side pipe. The assembly may include an engine, such as a horizontal shaft engine. In this embodiment, the horizontal shaft engine may be arranged on a first surface of

the platform and the self-winding hose reel may be arranged on a second surface of the platform and below the horizontal shaft engine.

(77) Thus, assemblies for high-pressure cleaning and methods for manufacturing an assembly for high-pressure cleaning have been described. Although embodiments have been described with reference to specific example embodiments, it will be evident that various modifications and changes can be made to these example embodiments without departing from the broader spirit and scope of the present application. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense.

Claims

1. An assembly for high-pressure cleaning, the assembly comprising: a self-winding hose reel; a pressure washer including a pump and a frame; and a fastening mechanism designed to secure the self-winding hose reel to the frame; wherein: the frame includes a first side wheel and a second side wheel parallel to the first side wheel; the frame includes a platform arranged between the first side wheel and the second side wheel; the assembly further comprises an engine, the engine being arranged on a first surface of the platform; the self-winding hose reel is arranged on a second surface of the platform and below the engine; and the self-winding hose reel is secured to the frame between the first side wheel and the second side wheel.
 2. The assembly of claim 1, wherein the pressure washer includes one of the following: an electrically powered high-pressure washer and a gasoline engine powered high-pressure washer.
 3. A method for manufacturing an assembly for high-pressure cleaning, the method comprising: providing a self-winding hose reel; providing a pressure washer including a pump and a frame; and providing a fastening mechanism designed to secure the self-winding hose reel to the frame; wherein: the frame includes a first side wheel and a second side wheel parallel to the first side wheel; the frame includes a platform arranged between the first side wheel and the second side wheel; the assembly further comprises an engine, the engine being arranged on a first surface of the platform; the self-winding hose reel is arranged on a second surface of the platform and below the engine; and the self-winding hose reel is secured to the frame between the first side wheel and the second side wheel.
 4. The method of claim 3, wherein the pressure washer includes one of the following: an electrically powered high-pressure washer and a gasoline engine powered high-pressure washer.
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